

MCQ Answer Generator

Sure! Paste the MCQs, and I'll format them with both the question and the correct answer clearly mentioned.

PART-A (AIML MCQs with Answers)

1.1 What is the primary advantage of depth-first search (DFS) over breadth-first search?

Answer: DFS requires less memory compared to BFS.

1.2 Differentiate between a rational agent and a non-rational agent.

Answer:

- **Rational Agent:** Acts in a way that maximizes its performance based on the available knowledge.
- **Non-Rational Agent:** Acts without considering the best possible outcome.

1.3 A search considered optimal when the heuristic is admissible. Give reason.*

Answer: An **admissible heuristic** never overestimates the true cost, ensuring that A* always finds the optimal solution.

1.4 In a two-player game, how does the minimax algorithm ensure that both players play optimally?

Answer: Minimax assumes both players play **optimally** by minimizing the possible loss for a worst-case scenario at each move.

1.5 How does simulated annealing differ from hill-climbing when reaching a local optimum?

Answer: Simulated annealing allows **occasional downhill moves**, helping to escape local optima, unlike hill-climbing.

1.6 In K-means, the objective is to minimize the within-cluster sum of squared distances.

Answer: Sum of squared distances between data points and their cluster centroids.

1.7 In Logistic Regression, the parameters are learned by minimizing the log-likelihood loss function, typically using gradient descent.

Answer: Log-likelihood loss function

1.8 Bagging improves the stability of a classifier by generating multiple datasets through bootstrap sampling and training a separate model on each dataset.

Answer: Bootstrap sampling

1.9 In a decision tree, the Gini index of a split is 0.5, and in another split, it is 0.3. Which split is considered better for classification? Give reason.

Answer: The split with Gini index 0.3 is better as a lower Gini index indicates higher purity of the split.

1.10 What are resubstitution errors?

Answer: Errors that occur when the training data itself is misclassified by the trained model.

1.11 When pruning a decision tree, why is it important to balance model complexity and accuracy?

Answer: To avoid overfitting and improve the model's generalization on unseen data.

1.12 Ensemble methods combine multiple individual models to create a stronger model that typically outperforms any individual model.

Answer: Stronger model

1.13 In a Naive Bayes classifier, the likelihood of observing a feature value given a class is calculated using conditional probability.

Answer: Conditional probability

1.14 A major disadvantage of the Nearest Neighbor Classifier is that it can become computationally expensive as the number of data points in the dataset increases.

Answer: Data points

1.15 To improve the accuracy of K-means clustering, it may be helpful to apply data normalization to the dataset before clustering to normalize the data.

Answer: Data normalization

✅ This format is suitable for direct pasting with both questions and answers!

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Answer Any FIVE Full Questions (Each Carries 20 Marks)

1.1 What does an agent's choice of action at any given instant depend on?

Answer: An agent's action depends on its perception of the environment, its knowledge base, and its decision-making algorithm.

1.2 What are the factors that are considered for judging the Search algorithms?

Answer: The key factors are:

- **Completeness** (Does it guarantee a solution if one exists?)
- **Optimality** (Does it find the best solution?)
- **Time complexity** (How much time does it take?)
- **Space complexity** (How much memory does it use?)

1.3 Give the PEAS description for a garbage-picking robot.

Answer:

- **P (Performance Measure):** Amount of garbage collected, efficiency, battery usage
- **E (Environment):** Streets, bins, trash locations, obstacles
- **A (Actuators):** Robot arm, wheels, sensors
- **S (Sensors):** Camera, proximity sensor, GPS

1.4 An inference algorithm that derives only entailed sentences is called ____.

Answer: Sound inference algorithm

1.5 In the real world, most of the constraint satisfaction problems have ____ domains.

Answer: Infinite or large domains

1.6 Represent "Gold and silver ornaments are precious" in First Order Logic.

Answer: $\forall x (G(x) \vee S(x) \rightarrow P(x))$

1.7 Categories of ____ are often characterized by structural relations among parts.

Answer: Objects

1.8 What is a deterministic node?

Answer: A **deterministic node** is a node in a decision tree where the outcome is **fully predictable**, meaning that given an input, there is only one possible output.

1.9 What is the input and output of the Gibbs sampling algorithm?

Answer:

- **Input:** Conditional probability distribution and an initial state
- **Output:** A sequence of samples that approximate the target distribution

1.10 What value is used to adjust the weights a small amount in the direction that reduces the error on the training examples?

Answer: Learning rate (η)

1.11 The _____ characterizes the (im)purity of an arbitrary collection of examples.

Answer: Entropy (or Gini Index in decision trees)

1.12 Consider a robot driving learning problem. What is the Performance measure and Training experience?

Answer:

- **Performance Measure:** Safety, fuel efficiency, speed limit adherence
- **Training Experience:** Past driving data, sensor readings, real-world driving feedback

1.13 Bring out the significance of the word “local” in locally weighted regression.

Answer:

Answer: The term “local” means that predictions are made using **only nearby training points**, giving more weight to closer points and ignoring far ones.

1.14 Give one disadvantage of instance-based approaches.

Answer:

High computational cost, as they require storing and comparing all training instances during prediction.

1.15 In reinforcement learning, under what condition does the agent prefer state 1 over state 2? Give reason for the preference.

Answer: The agent prefers **State 1 over State 2** if it has a **higher expected reward** based on the **value function** or **Q-value**. The preference is based on maximizing long-term rewards.

 This format ensures both the question and correct answer are well-structured for easy reference.

Answers

1.1 Probability of having meningitis given a stiff neck:

Using Bayes' Theorem:

$$P(M|S) = \frac{P(S|M)P(M)}{P(S)}$$

Given:

- $P(S|M) = 0.5$ (Probability of stiff neck given meningitis)
- $P(M) = \frac{1}{50000}$ (Probability of meningitis)
- $P(S) = \frac{1}{20}$ (Probability of stiff neck)

$$P(M|S) = \frac{0.5 \times \frac{1}{50000}}{\frac{1}{20}} = \frac{0.5}{50000} \times 20 = \frac{10}{50000} = 0.0002 = 0.02\%$$

Answer: 0.02% of people with a stiff neck have meningitis.

1.2 Two disadvantages of instance-based learning:

1. **High storage requirement** – Needs to store all training examples.
2. **Slow during prediction** – Must compare new data to all stored instances.

1.3 Bayesian Network Independence:

- a) D and E are not necessarily independent given A and B because they may still share dependencies through other variables.
- b) A and H are not necessarily independent given C because C may not fully separate their influence.

1.4 Modus Ponens Rule:

If $P \rightarrow Q$ (If P then Q) and P is true, then we conclude Q is true.

1.5 Informed Search Algorithm:

Search strategies that use **heuristic information** to guide the search towards the goal more efficiently.

1.6 Algorithm used in Game Trees for decision-making:

Minimax Algorithm

1.7 AI technique for understanding associations between objects and events:

Knowledge Representation and Reasoning (KRR)

1.8 Time and Space Complexity of BFS:

- **Time Complexity:** $O(b^d)$ (where b = branching factor, d = depth)
- **Space Complexity:** $O(b^d)$ (stores all nodes in memory)

1.9 AI term for common sense in problem-solving:

Heuristic Knowledge

1.10 Search algorithm that removes unnecessary branches:

Alpha-Beta Pruning

1.11 Process that makes two logical expressions identical:

Unification

1.12 Feature Type for Student Grades (A, B, C, D, E, F):

Ordinal Feature (because grades have an inherent order)

1.13 Four types of Informed Search Algorithms:

1. Greedy Best-First Search
2. *A Search**
3. *Iterative Deepening A (IDA)***
4. *Weighted A Search**

1.14 Classifying reviews as positive, negative, or neutral is an example of:

Sentiment Analysis (Supervised Learning)

1.15 Impact of high variance on the training set:

Overfitting (Model performs well on training data but poorly on new data)

✓ This format ensures each question is paired with its correct answer for easy reference.



Here are some multiple-choice questions (MCQs) based on the syllabus:

Unit I: Introduction, Intelligent Agents & Uninformed Search

1. What is the primary goal of Artificial Intelligence (AI)?
 - a) To create machines that act randomly
 - b) To build systems that mimic human intelligence

c) To replace humans in all tasks

d) To create algorithms without logical reasoning

✓ Answer: b) To build systems that mimic human intelligence

2. Which of the following is NOT a characteristic of an intelligent agent?

a) Autonomy

b) Reactivity

c) Randomness

d) Learning

✓ Answer: c) Randomness

3. Which search strategy explores all nodes at the current depth before moving to the next level?

a) Depth-First Search

b) Iterative Deepening Search

c) Breadth-First Search

d) A* Search

✓ Answer: c) Breadth-First Search

4. What is the major drawback of Depth-First Search (DFS)?

a) High memory consumption

b) Inefficiency in finding the shortest path

c) Requires a heuristic function

d) Always finds the optimal solution

✓ Answer: b) Inefficiency in finding the shortest path

5. Which search technique is a combination of DFS and BFS?

a) Iterative Deepening DFS

b) Best-First Search

c) Hill Climbing

d) A* Algorithm

✓ Answer: a) Iterative Deepening DFS

Unit II: Heuristic Search & Adversarial Search

6. What is the key advantage of A Search over other search algorithms?*

a) It requires no memory

- b) It guarantees finding the optimal solution
- c) It uses random heuristics
- d) It does not use a cost function

✓ Answer: b) It guarantees finding the optimal solution

7. Which search algorithm uses a fitness function to evolve solutions?

- a) Hill Climbing
- b) Genetic Algorithm
- c) Simulated Annealing
- d) A* Search

✓ Answer: b) Genetic Algorithm

8. Alpha-Beta pruning is used in which type of AI problem?

- a) Constraint Satisfaction
- b) Learning Problems
- c) Adversarial Search
- d) Uninformed Search

✓ Answer: c) Adversarial Search

9. What is the main limitation of Hill Climbing?

- a) It uses too much memory
- b) It can get stuck in local maxima
- c) It is an uninformed search strategy
- d) It does not work for optimization problems

✓ Answer: b) It can get stuck in local maxima

10. Simulated Annealing helps overcome which limitation of Hill Climbing?

- a) High memory usage
- b) Getting stuck in local optima
- c) Finding the shortest path
- d) Reducing heuristic functions

✓ Answer: b) Getting stuck in local optima

Unit III: Supervised Learning & Decision Trees

11. Which of the following is a supervised learning algorithm?

- a) K-means Clustering

- b) A* Search
- c) Decision Tree
- d) Alpha-Beta Pruning

✓ Answer: c) Decision Tree

12. Which problem occurs when a model performs well on training data but poorly on test data?

- a) Underfitting
- b) Overfitting
- c) Generalization
- d) Optimization

✓ Answer: b) Overfitting

13. What is the primary use of a validation set in model selection?

- a) To train the model
- b) To evaluate unseen data
- c) To improve overfitting
- d) To measure training error

✓ Answer: b) To evaluate unseen data

14. Which measure is commonly used to select the best attribute in a Decision Tree?

- a) Random Selection
- b) Heuristic Search
- c) Information Gain
- d) Path Cost

✓ Answer: c) Information Gain

15. Why is pruning used in Decision Trees?

- a) To increase model complexity
- b) To reduce overfitting
- c) To remove important nodes
- d) To increase training time

✓ Answer: b) To reduce overfitting

Unit IV: Classification Techniques & Ensemble Methods

16. What is the assumption in Naïve Bayes classification?

- a) Features are dependent
- b) Features are conditionally independent
- c) Features have no impact on classification
- d) Features require supervised learning

✓ Answer: b) Features are conditionally independent

17. Which classification technique is based on the closest training examples?

- a) Naïve Bayes
- b) Nearest Neighbor Classifier
- c) Logistic Regression
- d) Decision Tree

✓ Answer: b) Nearest Neighbor Classifier

18. Which ensemble method combines multiple weak classifiers to create a strong classifier?

- a) K-means
- b) Bagging
- c) Genetic Algorithm
- d) Gradient Descent

✓ Answer: b) Bagging

19. Logistic Regression is mainly used for:

- a) Classification problems
- b) Clustering problems
- c) Pathfinding problems
- d) Game Playing

✓ Answer: a) Classification problems

20. Random Forest is an ensemble learning method that consists of multiple:

- a) Neural Networks
- b) Clusters
- c) Decision Trees
- d) Heuristic Functions

✓ Answer: c) Decision Trees

Unit V: Unsupervised Learning & Clustering

21. Which algorithm is primarily used for clustering?

- a) A* Search
- b) K-means
- c) Logistic Regression
- d) Naïve Bayes

✓ Answer: b) K-means

22. Which clustering method divides data into k groups based on similarity?

- a) Decision Tree
- b) Genetic Algorithm
- c) K-means Clustering
- d) Alpha-Beta Pruning

✓ Answer: c) K-means Clustering

23. What is a key challenge in K-means clustering?

- a) Choosing the correct number of clusters
- b) Finding the shortest path
- c) Defining heuristic functions
- d) Ensuring all data is labeled

✓ Answer: a) Choosing the correct number of clusters

24. Which measure is used to evaluate clustering performance?

- a) Cohesion and Separation
- b) Information Gain
- c) Entropy
- d) Backpropagation

✓ Answer: a) Cohesion and Separation

25. Which clustering algorithm repeatedly divides data into two clusters?

- a) K-means
- b) Bisecting K-means
- c) Logistic Regression
- d) Nearest Neighbor

✓ Answer: b) Bisecting K-means

Final Thoughts

These MCQs cover AI fundamentals, search strategies, learning algorithms, classification techniques, and clustering as per your syllabus. Let me know if you need more! 🚀

Unit I: Introduction, Intelligent Agents & Uninformed Search

Introduction to AI

1. What is Artificial Intelligence (AI)?

Answer: AI is the simulation of human intelligence in machines that can perform tasks like reasoning, learning, and problem-solving.

2. What are the primary goals of AI?

Answer: The goals include learning, reasoning, problem-solving, perception, and language understanding.

3. Which fields contribute to AI development?

Answer: Mathematics, Computer Science, Neuroscience, Psychology, and Linguistics.

4. What is the Turing Test?

Answer: A test proposed by Alan Turing to determine if a machine can exhibit human-like intelligence.

5. What are the different types of AI?

Answer: Narrow AI, General AI, and Super AI.

Intelligent Agents

6. What is an intelligent agent?

Answer: An entity that perceives its environment and takes actions to maximize its success.

7. What are the key components of an intelligent agent?

Answer: Sensors, Actuators, and a Decision-Making Unit.

8. What is the difference between a rational agent and an omniscient agent?

Answer: A rational agent acts to maximize expected performance based on available information, while an omniscient agent knows the outcome of all actions beforehand.

9. What are the different types of intelligent agents?

Answer: Simple Reflex Agents, Model-Based Reflex Agents, Goal-Based Agents, Utility-

Based Agents, and Learning Agents.

10. What are the four types of agent environments?

Answer: Fully observable vs. partially observable, Deterministic vs. stochastic, Static vs. dynamic, Discrete vs. continuous.

Problem Solving & Uninformed Search Strategies

11. What is a problem-solving agent?

Answer: An agent that decides what actions to take by searching for sequences of actions that lead to desirable states.

12. What are the components of a well-defined problem in AI?

Answer: Initial state, Goal state, and Actions.

13. What is an uninformed search strategy?

Answer: A search strategy that explores the search space without additional knowledge about the goal's location.

14. How does Breadth-First Search (BFS) work?

Answer: It explores all nodes at the current depth level before moving to the next level.

15. What is the main disadvantage of BFS?

Answer: High memory usage due to storing all nodes at each level.

16. How does Depth-First Search (DFS) work?

Answer: It explores as deep as possible along each branch before backtracking.

17. What is the main limitation of DFS?

Answer: It may get stuck in infinite loops if the search space is large or infinite.

18. What is Depth-Limited Search (DLS)?

Answer: A variation of DFS that limits the depth of exploration to avoid infinite loops.

19. What is Iterative Deepening Depth-First Search (IDDFS)?

Answer: It repeatedly applies DFS with increasing depth limits to combine the benefits of DFS and BFS.

20. What is the advantage of IDDFS over DFS and BFS?

Answer: It finds the shortest path like BFS while using memory efficiently like DFS.

Unit II: Informed Search, Local Search & Adversarial Search

Informed (Heuristic) Search Strategies

1. What is an informed search strategy?

Answer: A search that uses problem-specific knowledge (heuristics) to find a solution more efficiently than uninformed search.

2. What is a heuristic function?

Answer: A function that estimates the cost of reaching the goal from a given state.

3. What is A search?*

Answer: A search algorithm that uses the function $f(n) = g(n) + h(n)$, where $g(n)$ is the cost from the start node to n , and $h(n)$ is the estimated cost from n to the goal.

4. Why is A search considered optimal and complete?*

Answer: It guarantees the shortest path when the heuristic function is admissible (never overestimates the cost).

5. What is the difference between Greedy Best-First Search and A Search?*

Answer: Greedy Best-First Search uses only $h(n)$, while A* uses both $g(n)$ and $h(n)$.

Beyond Classical Search: Local Search & Optimization

6. What is Local Search?

Answer: A search technique that explores neighboring solutions to find an optimal solution rather than keeping a full search tree.

7. What is the main advantage of Local Search?

Answer: It requires less memory and is effective for large or infinite state spaces.

8. How does Hill-Climbing search work?

Answer: It continuously moves to the neighboring state with the highest value, stopping at a local maximum.

9. What is the problem with Hill-Climbing?

Answer: It can get stuck in local maxima, plateaus, and ridges.

10. What is Simulated Annealing?

Answer: A probabilistic search that avoids local maxima by allowing bad moves with decreasing probability over time.

11. What is the role of the temperature parameter in Simulated Annealing?

Answer: It controls the probability of accepting worse solutions, decreasing over time to refine the search.

12. What is Local Beam Search?

Answer: It keeps track of multiple candidate states and selects the best k successors at each step.

13. How does Genetic Algorithm (GA) work?

Answer: It evolves solutions using selection, crossover, and mutation operations inspired by natural evolution.

14. What is the fitness function in Genetic Algorithms?

Answer: It evaluates how good a solution is in solving the problem.

15. What is the purpose of mutation in Genetic Algorithms?

Answer: It introduces diversity to prevent premature convergence.

Adversarial Search: Game Playing & Decision Making

16. What is an adversarial search?

Answer: A search where two or more agents compete, such as in games like chess or tic-tac-toe.

17. What is the Minimax algorithm?

Answer: A decision-making algorithm that minimizes the possible loss for a worst-case scenario in two-player games.

18. How does Minimax evaluate a game tree?

Answer: It assigns values to terminal states and propagates the best values up the tree.

19. What is the main limitation of Minimax?

Answer: It is computationally expensive for large search trees.

20. What is Alpha-Beta Pruning?

Answer: An optimization technique that eliminates branches in the game tree that do not affect the final decision.

21. **What is the advantage of Alpha-Beta Pruning?**

Answer: It reduces the number of nodes evaluated, making Minimax more efficient.

22. **What is a cutoff test in game search?**

Answer: A condition that stops expanding a search tree at a certain depth for practical efficiency.

23. **What is an evaluation function in games?**

Answer: A function that estimates the desirability of a game state when a full search is impractical.

24. **What type of AI is used in modern game-playing systems like Deep Blue?**

Answer: A combination of Minimax, Alpha-Beta Pruning, and Machine Learning techniques.

25. **What is the significance of Monte Carlo Tree Search (MCTS)?**

Answer: It is an alternative to Minimax used in games like Go, relying on random sampling of the game tree.

This covers **Unit II** in depth. Let me know if you want **Unit III** next! 🚀

Unit III: Supervised Learning & Decision Trees

Supervised Learning: Basic Concepts

1. **What is supervised learning?**

Answer: A type of machine learning where the model learns from labeled training data.

2. **What are the two main types of supervised learning?**

Answer: Classification and Regression.

3. **What is the goal of classification in supervised learning?**

Answer: To assign input data to predefined categories or labels.

4. **What is the goal of regression in supervised learning?**

Answer: To predict a continuous numerical value.

5. **What is the role of a loss function in supervised learning?**

Answer: It measures the error between the predicted and actual output.

Decision Tree Classifier

6. What is a decision tree?

Answer: A tree-like model used to make decisions based on feature values.

7. What is a node in a decision tree?

Answer: A point where a feature is tested to split data into branches.

8. What is a leaf node in a decision tree?

Answer: A terminal node representing the final class label or prediction.

9. What is entropy in decision trees?

Answer: A measure of the uncertainty or impurity in a dataset.

10. What is Information Gain in decision trees?

Answer: A measure of how much a feature reduces uncertainty when used for splitting.

11. Which algorithm is commonly used to build decision trees?

Answer: ID3 (Iterative Dichotomiser 3).

12. What is the role of the Gini index in decision trees?

Answer: It is an impurity measure used to determine the best feature for splitting.

13. What is pruning in decision trees?

Answer: The process of removing branches to prevent overfitting.

14. What are the advantages of decision trees?

Answer: They are easy to interpret and handle both numerical and categorical data.

15. What is a disadvantage of decision trees?

Answer: They are prone to overfitting, especially with complex data.

Model Overfitting & Selection

16. What is model overfitting?

Answer: When a model learns noise in the training data instead of generalizing well.

17. What is the main reason for overfitting?

Answer: Having too many features or a complex model.

18. How can overfitting be reduced?

Answer: Using regularization, pruning, or collecting more data.

19. **What is a validation set used for?**

Answer: To fine-tune the model and prevent overfitting.

20. **What is cross-validation?**

Answer: A technique that splits data into multiple parts to train and test the model iteratively.

21. **What is the difference between training and testing datasets?**

Answer: The training set is used to train the model, while the testing set evaluates its performance.

22. **What is model complexity?**

Answer: A measure of how detailed or flexible a model is in learning patterns.

23. **Why is simpler model selection preferred in machine learning?**

Answer: To reduce overfitting and improve generalization to new data.

24. **What is the significance of statistical bounds in model selection?**

Answer: They help determine the expected accuracy of a model.

25. **How is model evaluation done in supervised learning?**

Answer: Using metrics like accuracy, precision, recall, and F1-score.

This covers **Unit III** in depth! Let me know if you want **Unit IV** next. 🚀

Unit IV: Nearest Neighbor, Naïve Bayes, Logistic Regression, and Ensemble Methods

Nearest Neighbor Classifiers

1. **What is the nearest neighbor classifier?**

Answer: A classification algorithm that assigns a class to a data point based on the majority class of its nearest neighbors.

2. **What is the most common distance metric used in the k-Nearest Neighbors (k-NN) algorithm?**

Answer: Euclidean distance.

3. How does the value of 'k' affect the performance of k-NN?

Answer: A small k makes the model sensitive to noise, while a large k smooths decision boundaries.

4. What is the main disadvantage of the k-NN algorithm?

Answer: It is computationally expensive because it requires storing and comparing all training data.

5. Is k-NN a parametric or non-parametric algorithm?

Answer: Non-parametric.

Naïve Bayes Classifier

6. What assumption does the Naïve Bayes classifier make?

Answer: It assumes that all features are independent given the class label.

7. What is the formula for Bayes' theorem?

Answer: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

8. Why is Naïve Bayes called "naïve"?

Answer: Because it assumes conditional independence between features, which is often not true in real-world data.

9. What type of probability distribution does the Gaussian Naïve Bayes classifier assume for continuous data?

Answer: A normal (Gaussian) distribution.

10. What is an advantage of the Naïve Bayes classifier?

Answer: It is computationally efficient and works well with small datasets.

Logistic Regression

11. What type of problem is logistic regression mainly used for?

Answer: Binary classification problems.

12. What is the output range of logistic regression?

Answer: Between 0 and 1.

13. **Which function is used in logistic regression to transform linear outputs?**
Answer: The sigmoid function.
 14. **How is the decision boundary determined in logistic regression?**
Answer: By setting a threshold (usually 0.5) on the predicted probability.
 15. **What is the cost function used in logistic regression?**
Answer: Log loss or cross-entropy loss.
 16. **What optimization technique is commonly used for training logistic regression?**
Answer: Gradient Descent.
 17. **What happens if we use logistic regression for a multi-class classification problem?**
Answer: We use extensions like One-vs-All (OvA) or Softmax regression.
 18. **What is regularization in logistic regression?**
Answer: A technique to prevent overfitting by adding a penalty term (L1 or L2) to the cost function.
 19. **Which regularization technique uses absolute values of coefficients?**
Answer: Lasso (L1) regularization.
 20. **Which regularization technique uses squared values of coefficients?**
Answer: Ridge (L2) regularization.
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Ensemble Methods

21. **What is an ensemble method?**
Answer: A technique that combines multiple models to improve performance.
22. **What is bagging in ensemble learning?**
Answer: A method that trains multiple models on random subsets of data and averages their predictions.
23. **Which algorithm is an example of bagging?**
Answer: Random Forest.
24. **What is boosting in ensemble learning?**
Answer: A method that trains models sequentially, where each new model corrects the errors of the previous ones.

25. Which algorithm is an example of boosting?

Answer: AdaBoost or Gradient Boosting.

26. How does Random Forest improve decision trees?

Answer: By using multiple decision trees trained on random subsets of data and features.

27. What is the key advantage of ensemble methods?

Answer: They reduce overfitting and improve generalization.

28. What is a weakness of ensemble methods?

Answer: They can be computationally expensive and difficult to interpret.

29. Which ensemble method gives more importance to difficult cases?

Answer: Boosting.

30. What is the difference between hard and soft voting in ensemble learning?

Answer: Hard voting selects the most common class, while soft voting averages predicted probabilities.

That covers **Unit IV** comprehensively! Let me know if you want **Unit V** next. 🚀

Unit V: Unsupervised Learning & Clustering

Unsupervised Learning

1. What is unsupervised learning?

Answer: A type of machine learning where the model learns patterns from unlabeled data.

2. What is the main goal of clustering in unsupervised learning?

Answer: To group similar data points together based on their features.

3. Which type of learning does clustering belong to?

Answer: Unsupervised learning.

4. Name a real-world application of clustering.

Answer: Customer segmentation in marketing.

5. What is the main challenge in unsupervised learning?

Answer: Evaluating the quality of clusters without ground truth labels.

Types of Clustering

6. What are the two main types of clustering methods?

Answer: Hierarchical clustering and partition-based clustering.

7. Which clustering method builds a tree-like structure of clusters?

Answer: Hierarchical clustering.

8. What is the main difference between hard and soft clustering?

Answer: Hard clustering assigns each data point to a single cluster, while soft clustering assigns probabilities of belonging to multiple clusters.

9. What is an example of soft clustering?

Answer: Fuzzy C-Means (FCM).

10. Which clustering method divides data into k predefined clusters?

Answer: K-Means clustering.

K-Means Clustering

11. What is the main assumption of K-Means clustering?

Answer: Data is grouped into spherical clusters with equal variance.

12. What is the first step in the K-Means algorithm?

Answer: Initialize k cluster centroids randomly.

13. Which metric is commonly used in K-Means clustering to assign data points to clusters?

Answer: Euclidean distance.

14. What happens if k is chosen too small in K-Means clustering?

Answer: Some distinct groups may be merged into one cluster.

15. What happens if k is chosen too large in K-Means clustering?

Answer: Similar clusters may be split unnecessarily.

16. **What is a common method to determine the optimal number of clusters in K-Means?**
Answer: The Elbow Method.
 17. **What is a limitation of K-Means clustering?**
Answer: It assumes clusters are spherical and equally sized.
 18. **Which variation of K-Means splits clusters into two at each step?**
Answer: Bisecting K-Means.
 19. **What is a major drawback of K-Means clustering?**
Answer: It is sensitive to initial centroid selection.
 20. **Which technique is used to improve the initialization of K-Means?**
Answer: K-Means++.
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Cluster Evaluation

21. **Why is cluster evaluation important?**
Answer: To measure the quality of clusters and determine the best clustering approach.
22. **Which two main types of cluster evaluation exist?**
Answer: Internal and external evaluation.
23. **What does internal cluster evaluation measure?**
Answer: The cohesion and separation of clusters without using external labels.
24. **What does external cluster evaluation measure?**
Answer: The similarity between clustering results and ground truth labels.
25. **What metric is commonly used for internal cluster evaluation?**
Answer: Silhouette coefficient.
26. **What does a high Silhouette coefficient indicate?**
Answer: Well-separated and well-defined clusters.
27. **What is the main challenge in determining the number of clusters?**
Answer: There is no universally correct number; it depends on the dataset.
28. **Which technique is used to estimate the best number of clusters based on variance?**
Answer: The Elbow Method.
29. **What does a proximity matrix in clustering represent?**
Answer: The pairwise distance between data points.

30. Why is choosing the right clustering validity measure important?

Answer: It ensures the quality and reliability of the clustering results.

That completes **Unit V** with deep●